ALGEBRA 1 · QUADRATIC FUNCTIONS AND EQUATIONS

Parabola Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Parabola Explorer** on Mr. K's Classroom: Algebra 1 → Quadratic Functions and Equations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict

before you touch the tool

1. What do you think a negative value of a does to $y = ax^2$?

2. Compared with $y = x^2$, which way does $y = (x - 3)^2$ move?

2 Explore

use the tool and record what you see

Vertex form $y = a(x - h)^2 + k$

Equation	Vertex	Opens	Compared with $y = x^2$
$y = x^2 + 4$			
$y = (x - 3)^2$			
$y = (x + 2)^2 - 5$			
$y = -x^2$			
$y = 3x^2$			
$y = \frac{1}{2}(x - 1)^2 + 2$			

Standard form $y = ax^2 + bx + c$

Use the Standard form tab.

Equation	Discriminant $b^2 - 4ac$	Number of x-intercepts
$y = x^2 - 2x - 3$		
$y = x^2 - 4x + 4$		
$y = x^2 + 2x + 5$		
$y = -x^2 + 6x - 5$		

3 Write the rule

fill in each blank from what you observed

1. k moves the parabola _____ or _____.

2. h moves the parabola left or right, in the _____ direction of its sign in $(x - h)$.

3. If $a < 0$ the parabola opens _____. If $|a| > 1$ it is _____ than $y = x^2$.

4. The vertex of $y = a(x - h)^2 + k$ is _____.

5. If $b^2 - 4ac > 0$ there are _____ x-intercepts; if it equals 0 there is _____; if it is negative there are _____.

4 Check no tool: show your work

1. Give the vertex of $y = -2(x + 4)^2 + 1$. Which way does it open?

2. Write $y = x^2$ shifted left 3 and down 6.

3. How many x-intercepts does $y = 2x^2 + 3x + 5$ have?

4. Find the axis of symmetry of $y = x^2 - 8x + 3$.

5 Reflect

Why does $(x - 3)^2$ move the graph right, not left?